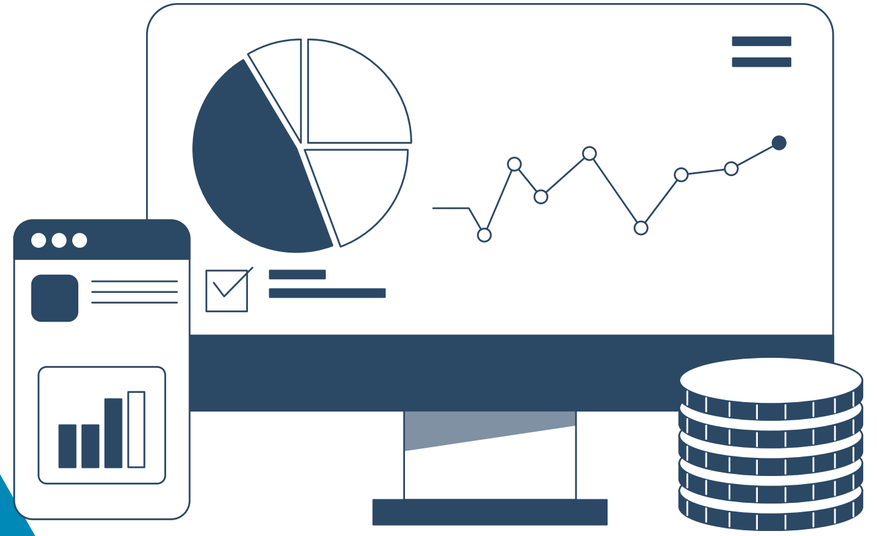
DATA ANALYTICS

30 days



DAY 21

Agenda

- Why Queries Become Slow
- Index
- Composite Index
- Query Optimization
- EXPLAIN Statement
- SQL Performance Best Practices

Why Queries Become Slow?

A query becomes slow when the database has to process a large amount of data before finding the required result.

- Full Table Scan
- Missing Index
- Using SELECT *
- Retrieving unnecessary rows
- Inefficient JOIN operations
- Using leading wildcards ('%text')
- Large tables with millions of records

What is an Index?

An Index is a special data structure that helps the database find data quickly without scanning the entire table.Think of it like the index page of a book.

```
CREATE INDEX idx_city
ON Sales(City);

SELECT *
FROM Sales
WHERE City='Dhaka';
```

What is a Composite Index?

A Composite Index is an index created using multiple columns.

It is useful when queries frequently filter using more than one column.

```
CREATE INDEX idx_city_customer
ON Sales(City, CustomerName);

SELECT *
FROM Sales
WHERE City='Dhaka'
AND CustomerName='Rahim';
```

What is Query Optimization?

SQL Performance Optimization is the process of making queries run faster by reducing unnecessary data scanning and helping the database find data more efficiently.

Goals:

- Reduce scanned rows
- Reduce execution time
- Improve database performance

What is the EXPLAIN Statement?

The EXPLAIN statement shows how the database executes a query.

It helps identify:

- Whether an index is used
- Number of rows scanned
- Query execution strategy
- Performance bottlenecks

SQL Performance Best Practices

- Create indexes on frequently searched columns
- Use Composite Indexes when filtering multiple columns
- Avoid SELECT *
- Use WHERE clauses to reduce scanned rows
- Use LIMIT when only a few records are needed
- Index JOIN columns
- Use EXPLAIN to analyze query performance
- Retrieve only necessary columns
- Avoid unnecessary calculations in queries
- Regularly review slow-running queries

THANK YOU

FOR YOUR ATTENTION